



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

CV1003-1 – ENGINEERING MECHANICS
 (BT, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

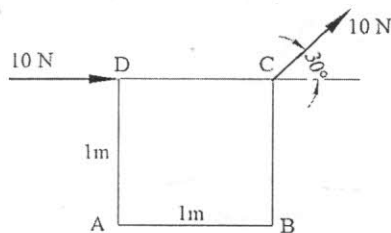
PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Two forces meeting at a point, but their line of action is not in one plane, is known as

A) Non-concurrent non-coplanar forces B) Non-concurrent coplanar forces
 C) Concurrent coplanar forces D) Concurrent non-coplanar forces

2. Moment with respect to the point C in the given figure is _____



A) 5 N-m
 B) 0 N-m
 C) 19.66 N-m
 D) 18.66 N-m

3. The Principle of Transmissibility of Forces states that, when a force acts upon a body, its effect is

A) Maximum if it acts at the centre of gravity of the body B) Different at different points on its line of action
 C) Same at every point on its line of action D) Minimum if it acts at the centre of gravity of the body

4. The effect of a force on a body depends upon its _____

A) Magnitude B) All the options
 C) Position D) Direction

5. What is the vertical centroidal distance from base of right angled triangle of base 20 cm and side 40 cm?

A) 13.33 cm B) 19.36 cm
 C) 13.28 cm D) 38.72 cm

6. If an area has one line of symmetry the centroid lie

A) Anywhere on the area B) Somewhere along the line symmetry
 C) Midpoint of the line of symmetry D) None of the above

7. What is the formula of radius of gyration?

A) $k^2 = I/A$ B) $k^2 = I^2/A$
 C) $k^2 = I^2/A^2$ D) $k^2 = (I/A)^{1/2}$

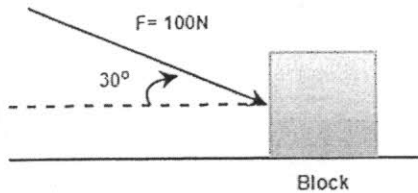
8. Moment of inertia of a triangle of base 2cm and height 3cm about its base is given by

A) 2.5cm^4 B) 2cm^4
 C) 4cm^4 D) 4.5cm^4

9. The frictional force is independent of

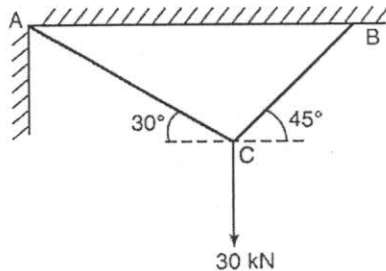
A) The coefficient of friction B) The area of contact
 C) The normal reaction D) The angle of friction

10. Force of 100N is exerted at an angle of 30° on the box kept on the floor with the coefficient of static friction of 0.2. Find the friction force.



- A) 36.6N
B) zero
C) 86.6N
D) 50N

11. A rigid body is in equilibrium if the resultant force of concurrent force system is
A) Negative
B) Positive
C) Zero
D) Maximum
12. The tension in cable BC to keep the body under equilibrium is

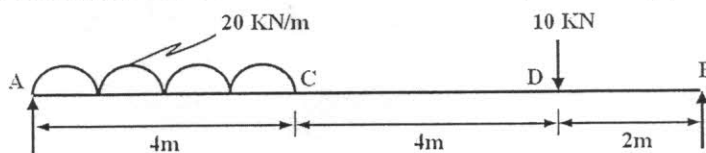


- A) 26.897 kN
B) 24.951 kN
C) 21.961 kN
D) 27.895 kN

13. A beam is a structural member which is subjected to

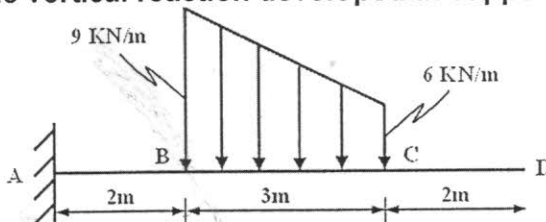
- A) Horizontal load
B) Axial tension or compression
C) Twisting moment
D) Transverse loads and couples
14. A beam which extends beyond its supports can be termed as _____
A) Propped Cantilever beam
B) Over hanging beam
C) Fixed beam
D) Continuous beam

15. Determine the vertical reaction developed at support B in the beam shown below,



- A) 90 kN
B) 16 kN
C) 66 kN
D) 24 kN

16. The vertical reaction developed at support A in the beam shown below is:



- A) 13.5 kN
B) 22.5 kN
C) 27 kN
D) 31.5 kN

17. At the point of contra-flexure, the value of bending moment is _____

- A) Maximum
B) minimum
C) Zero
D) Can't be determined

18. Shear force in a beam is

- A) Neither parallel nor perpendicular to the length
B) Perpendicular to the length
C) Parallel to the length
D) None

19. A simply supported beam of span 6 m carries uniformly distributed load of 1.5 kN/m, the maximum bending moment is

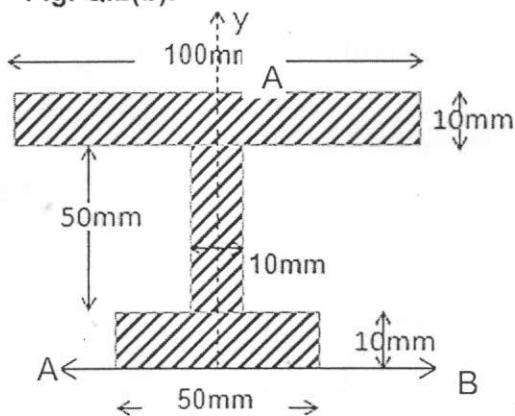
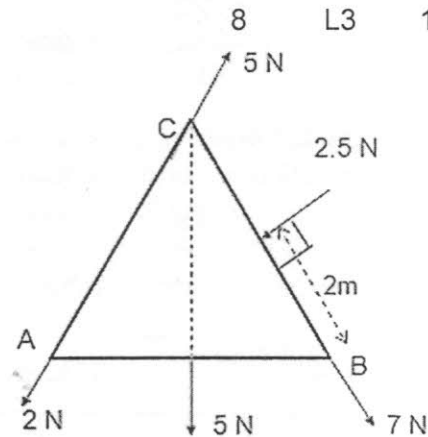
- A) 6.75 kN-m
B) 7.5 kN-m
C) 13 kN-m
D) 9 kN-m

20. A cantilever beam of length 2m carries a U.D.L. of 150 N/m over its whole span. The maximum shear force in the beam will be.

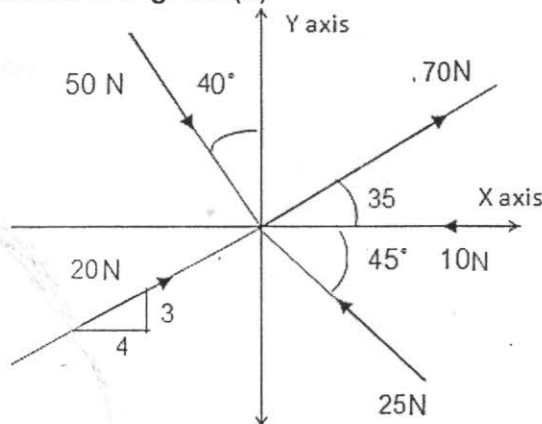
- A) 150 N
B) 150 N-m
C) 300 N
D) 300 N-m

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) State and explain principle of transmissibility of a force with neat sketch. State its limitations. | 8 | L*2 | 1 | 1,2 |
| b) Locate the centroid of shaded area with respect to given axes shown in Fig. Q.1(b). | 8 | L3 | 2 | 1,2 |
| 2. a) Derive an expression for centroid of a semi-circular section of radius 'r' by the method of integration. | 8 | L3 | 2 | 1,2 |
| b) Sketch the resultant of force system acting on the equilateral triangular plate element of sides 4m with respect to point B shown in Fig. Q.2(b). | 8 | L3 | 1 | 1,2 |

**Fig. Q.1(b)****Fig. Q.2(b)**

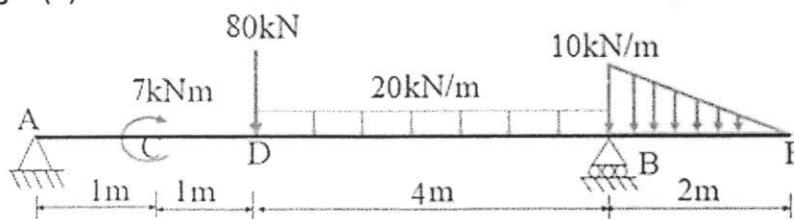
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|--|---|----|---|-----|
| 3. a) Distinguish centroid and centre of gravity. | 4 | L2 | 2 | 1,2 |
| b) Define the following i) polar moment of inertia ii) radius of gyration. | 4 | L1 | 2 | 1,2 |
| c) Determine the magnitude and direction of the resultant for the force system shown in Fig. Q.3(c). | | | | |

**Fig. 3(c)**

8	L3	1	1,2
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Unit – II

- | | | | | |
|---|---|----|---|-----|
| 4. a) State Coulomb's laws of friction (any four). | 4 | L1 | 3 | 1,2 |
| b) Explain free body diagram with an example. | 4 | L2 | 3 | 1,2 |
| c) Determine the supports reactions for the beam loaded shown in Fig. 4(c). | | | | |

**Fig. 4(c)**

8	L3	4	1,2
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5. a) Explain any four types of beams with neat sketches. 8 L2 4 1,2
 b) Determine the weight W_1 and W_2 such that string BC is horizontal shown in Fig. 5(b), Assume the pulley to be frictionless and mass less.

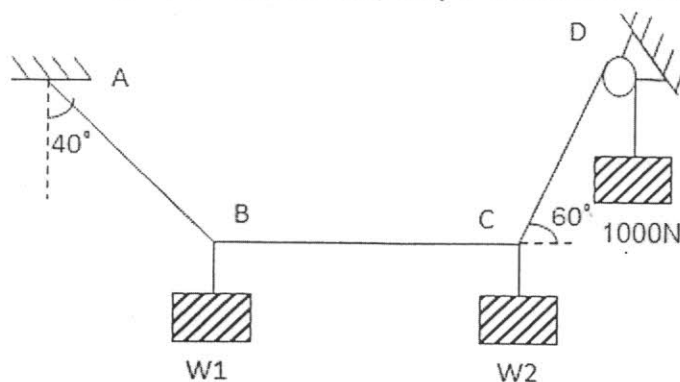


Fig. 5(b)

6. a) Define the following: i) angle of friction and ii) angle of repose. 8 L3 3 1,2
 b) Define equilibrium and mention the conditions of equilibrium for
 i) coplanar concurrent force system 4 L1 3 1,2
 ii) coplanar non-concurrent force system. 4 L2 3 1,2
 c) Determine the supports reactions for the beam loaded shown in Fig. 6(c).

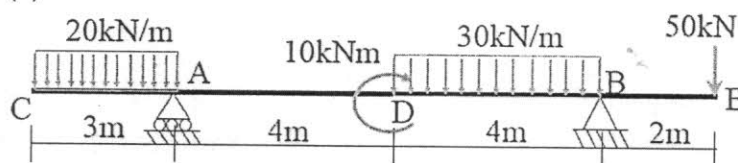


Fig. 6(c)

8 L3 4 1,2

Unit – III

7. a) Define shear force and bending moment diagram with sign convention. 4 L2 5 1,2
 b) Sketch the variation of shear force and bending moment for a loaded beam shown in Fig. 7(b) and find the contra flexure if any.

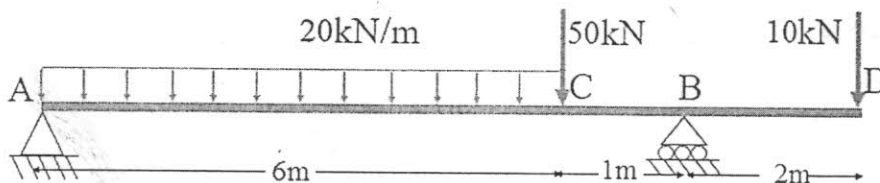


Fig. 7(b)

12 L3 5 1,2

8. a) Derive the relationship between intensity of loading, shear force and bending moment. 6 L3 5 1,2
 b) Sketch the variation of shear force and bending moment for a loaded beam shown in Fig. 8(b).

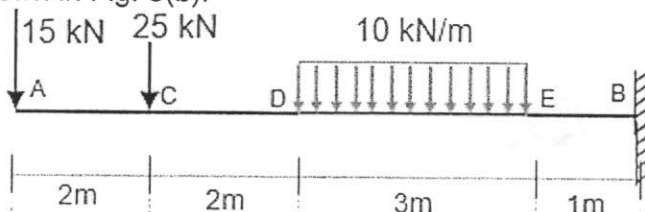


Fig. 8(b)

10 L3 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
